NNS in Stata: Implementation & Evaluation Plan

A strategic blueprint for bringing Nonlinear Nonparametric Statistics to Stata

Part 1: Foundations of NNS & Core Comparisons

1.1 What is NNS?

The Nonlinear Nonparametric Statistics (NNS) method, pioneered by Fred Viole and David Nawrocki, is a paradigm that relies on **Partial Moments** rather than Central Moments (mean, standard deviation, variance). By natively capturing variance strictly above or below a specific target point, NNS effortlessly accommodates asymmetric distributions without enforcing linear or symmetric assumptions.

1.2 The Functional Form

The fundamental equations driving NNS are the Lower and Upper Partial Moments:

$$LPM_n(t, X) = (1/N)\ \Sigma_{x_i \leq t}\ (t - x_i)^n$$

$$UPM_n(t, X) = (1/N)\ \Sigma_{x_i > t}\ (x_i - t)^n$$

In regression modeling, rather than applying a global mathematical function, NNS partitions the dataset dynamically. When the ratio of these partial moments shifts significantly (indicating a nonlinear inflection point), the algorithm creates a "knot" or partition. Predictions are then formed by averaging the dependent variable Y within these localized clusters.

1.3 NNS vs. Stata's `npregress`

Stata’s standard nonparametric regression (`npregress`) utilizes kernel methods or series/splines. A typical kernel regression applies distance-decaying weights to nearby points based on a bandwidth:

$$m(x) = \Sigma K((X_i - x)/h) Y_i / \Sigma K((X_i - x)/h)$$

Key Differentiators:

- **Bandwidth Selection:** `npregress` relies on computationally expensive cross-validation to find an optimal bandwidth h. NNS is *bandwidth-free*; intersections of LPM/UPM define partitions automatically.
- **Asymmetry:** Kernel methods often apply symmetric weighting (e.g., Gaussian kernels). NNS makes no such assumptions, adapting naturally to heavily skewed data.
- **Boundary Effects:** Kernel methods notoriously suffer at data edges (the boundary effect). NNS isolates extreme clusters naturally, handling boundaries without complex mathematical corrections.

Part 2: Stata Implementation Plan

2.1 Architecture and Approach

Writing a fast, robust NNS package for Stata requires utilizing **Mata** (Stata's matrix programming language) for the heavy computational lifting. Since NNS avoids iterative cross-validation, operations can be heavily vectorized.

- **Mata Core (nns_core.mata):** Will handle the sorting of vectors, calculation of LPMs/UPMs across empirical distributions, and the detection of partial-moment shift thresholds.
- **Stata Wrapper (nnsreg.ado):** A standard ado-file parsing Stata syntax. It will accept standard Stata commands like `nnsreg y x1 x2 [if] [in], [options]`.
- **Post-estimation:** Standard Stata features such as `predict` (for fitted values and out-of-sample extrapolation) and `margins` compatibility.

2.2 Comparisons and Differentiation in the Literature

To demonstrate value, the package must be explicitly compared against:

- **lpoly / npregress:** To highlight computational speed and lack of overfitting at boundaries.
- **mkspline (Cubic Splines):** Splines require the researcher to define the number of knots. NNS places knots automatically based on the data's native distribution.
- **Regression Discontinuity (rdrobust):** Show how NNS natively detects structural breaks/thresholds without forcing the researcher to manually specify the discontinuity cutoff point.

Part 3: Showcasing NNS in Applied Public Policy

While NNS has deep roots in finance (where downside risk vs. upside potential is highly asymmetric), it is extraordinarily well-suited for public policy evaluation where treatment effects are non-linear or feature severe threshold effects.

3.1 Environmental Policy & Public Health

Use Case: Evaluating the impact of particulate matter (PM2.5) on localized respiratory hospitalizations.

Why NNS: The biological response to pollution is rarely linear. There are threshold effects where health impacts spike exponentially after a certain concentration. NNS will automatically cluster and partition these thresholds without assuming a smooth polynomial curve, revealing the exact policy-relevant cutoffs where interventions are needed most.

3.2 Education Economics: Funding & Class Size

Use Case: Estimating the effect of per-pupil funding on standardized test scores.

Why NNS: Funding usually exhibits diminishing marginal returns (a highly asymmetrical relationship). NNS can identify the exact inflection point where additional funding stops yielding statistically meaningful test score improvements, providing policymakers with a clear "optimal funding" target rather than a smoothed, generalized curve.

3.3 Welfare and Tax Interventions

Use Case: Marginal Propensity to Consume (MPC) out of stimulus checks or welfare benefits across the income distribution.

Why NNS: Lower-income brackets have a completely different response profile to cash transfers than higher-income brackets. NNS will cluster the income distribution autonomously based on consumption shifts, providing a purely data-driven grouping of "high responders" vs. "low responders" without pre-defining income bins.

Part 4: Stata Journal Working Paper Skeleton

Title: *nnsreg: Nonlinear Nonparametric Statistics for Asymmetric and Boundary-Robust Modeling in Stata*

1. **Abstract:** Introduce NNS, highlighting its bandwidth-free nature and reliance on partial moments. State the problem with existing kernel-based smoothers in Stata (computational time, boundary issues) and how `nnsreg` solves them.
2. **Introduction:**
    - Brief history of NNS (Viole & Nawrocki, 2013).
    - The gap in applied microeconomics and public policy for fast, threshold-detecting nonparametric methods.
3. **Methodology (The Mathematics):**
    - Define LPM and UPM.
    - Explain the clustering/partitioning algorithm.
4. **Syntax and Options:**
    - Detailed breakdown of the `nnsreg` command structure, `predict` options, and plotting capabilities.
5. **Replication & Simulation (The Baseline):**
    - Replicate key nonlinear toy examples from Viole & Nawrocki (2013).
    - Monte Carlo simulation comparing computation time of `nnsreg` vs `npregress` on a dataset with N=10,000.
6. **Applied Public Policy Example:**
    - Walkthrough of an applied dataset (e.g., using a publicly available policy dataset like the NSW job training data or environmental air quality data).
    - Demonstrate how NNS finds policy-relevant thresholds organically.
7. **Conclusion:** Summary of benefits and potential extensions (e.g., integrating NNS causal tools).

Part 5: References & Resources

- Viole, F. and Nawrocki, D. (2013). *Nonlinear Nonparametric Statistics: Using Partial Moments*. ISBN: 1490523995. [Book Link]
- Viole, F., & Nawrocki, D. (2017). *Nonparametric Regression Using Clusters*. Computational Economics, 54, 759-776. DOI: 10.1007/s10614-017-9713-5
- NNS R Package (CRAN): https://cran.r-project.org/web/packages/NNS/
- NNS GitHub Repository: https://github.com/OVVO-Financial/NNS